Charvi Mehra

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Summary

AI/ML Engineer with production experience building multi-agent AI systems, LLM-powered platforms, and scalable ML infrastructure. Architected enterprise AI solutions serving tier-1 enterprises (PwC, Paytm, Blinkit) and open-source AI framework deployed across 2M+ websites. Strong foundation in distributed systems, async architecture, and edge ML deployment.Published research in applied machine learning as first author.

Skills

Languages: Python, SQL, R, C, PHP, Shell Scripting
AI/ML: LLM Integration, Multi-Agent Systems, RAG Pipelines, TensorFlow, PyTorch, LangChain, Computer Vision
Backend & Infrastructure: FastAPI, Docker, REST APIs, Async I/O, S3, Git
Data & Analytics: Pandas, NumPy, OpenCV, Statistical Modeling, Data Pipelines

Experience

AI Research Intern **Sep 2025 – Present**
Irame AI *Gurugram, India*

- Architected a **multi-agent AI feature** to automate enterprise audit report generation from large-scale, multi-source data; reduced turnaround time from 4+ hours to <60 seconds, deployed across **PwC, Paytm, Blinkit, and Havells**.
- Designed an agentic AI framework using **Planner–Actor pattern** with DAG-based task orchestration; achieved **5× throughput** via semaphore-limited parallel execution with thread-safe state management.
- Engineered an **async-first architecture** handling **1,000+ concurrent requests** using FastAPI, httpx, and S3; enabled automatic resource cleanup and full observability through per-execution trace logging.
- Reduced **LLM inference costs by 40–60%** via an optimized data embedding pipeline; built a dynamic report generation engine with Chart.js supporting **10+ interactive visualizations**.

Open Source Contributor **May 2025 – Aug 2025**
Google Summer of Code 2025, Joomla! *Remote*

- Shipped the **Joomla! AI Framework** to production, enabling AI capabilities across **2M+ active websites** through a provider-agnostic architecture supporting OpenAI, Anthropic, and Ollama backends.
- Built end-to-end generative AI features including content assistants, SEO optimization helpers, automated alt-text generation, and image generation with SQL-driven content retrieval ensuring data consistency.
- Authored technical documentation and standardized CI/CD pipelines, reducing contributor onboarding time by 50% and enabling production-grade deployment workflows for AI extensions.

Machine Learning Research Intern **Feb 2025 – Jul 2025**
CyPSi Lab, Institute of Informatics and Communication (IIC) *Delhi, India*

- **First-authored** a conference paper (accepted at InCoWoCo 2025) achieving **96.3% accuracy** on Indian currency classification with an ultra-compact **1.99 MB model** and **60 ms inference latency**.
- Designed optimized 4-block CNN architecture for mobile deployment; converted to **TensorFlow Lite** and integrated into Android application with real-time inference and Hindi speech output for accessibility.

Publications

[**C.1**] **Mehra, C.**, et al. (2025). An Ultra-Compact CNN for Real-Time Offline Indian Currency Classification on Mobile Devices. *Proceedings of the International Conference for Women in Computing (INCOWOCO 2025).*

[**J.1**] **Mehra, C.**, Kumar, P., & Kaushik, M. (2025). The marvels of DNA templated gold nanoclusters: Nature's miracle materials. *Next Materials*, Vol. 8, pp. 2949-8228. DOI: 10.1016/j.nxmate.2025.100665

Projects

Enterprise RAG Documentation Assistant | *Python, LangChain, FastAPI, Vector Databases*

- Built production RAG pipeline that **reduced documentation lookup time by 70%** for recurring analysis tasks by retrieving and summarizing technical reports from 10,000+ document knowledge bases.
- Implemented retrieval grounding validation layer ensuring responses are generated only from source documents, **eliminating hallucinations** in automated reporting workflows with 99.2% factual accuracy.

Automotive 3D Visualization Pipeline | *Python, 3D Gaussian Splatting, Diffusion Models, OpenCV*

- Engineered automotive visualization pipeline using **3D Gaussian Splatting** and multi-view geometry to generate interactive 360-degree vehicle renders from multi-angle image sets.
- Implemented latent diffusion-based editing for background replacement, surface cleanup, and lighting normalization, enabling **scalable marketing asset generation** reducing manual editing time by 80%.

Education

B.Tech. in Information Technology and Mathematical Innovation **Aug 2022 – Jun 2026**
Cluster Innovation Centre, University of Delhi *CGPA: 9.09/10*